

1. Hide a secret message inside an image

Using OpenStego

1. Install and open OpenStego.
2. Select Embed data.
3. Choose a normal image, for example:
 - original.jpg
4. Create a small text file:
 - secret.txt
5. Put something harmless inside it, for example:
 - This is a cybersecurity lab test message.
6. In OpenStego:
 - Message File: secret.txt
 - Cover File: original.jpg
 - Select an output filename such as:
 - stego.png
7. Click Embed Data.

Verify the image

- Open stego.png normally using Windows Photos or another image viewer.

2. Extract the hidden message

When your trainer gives you a stego image, for example:

- trainer_stego.png

Open OpenStego and select Extract data.

Choose:

- Input file: trainer_stego.png
- Output directory: extracted

If a password was used, enter the password supplied by your trainer.

After extraction, you should get something such as:

- extracted/secret.txt

Open it:

- This is the hidden message supplied for the lab.

3. Python script to clear logs.txt

First create:

- logs.txt

Example contents:

- 2026-09-17 User login successful
- 2026-09-17 Failed login attempt
- 2026-09-17 File accessed
- 2026-09-17 User logged out

Then create:

- clear_logs.py

Use:

```
with open("logs.txt", "w") as file:  
    pass  
  
print("logs.txt has been cleared.")
```

Run

- python clear_logs.py

The w mode truncates the existing file.

Check:

- Get-Content .\logs.txt

It should contain no log entries.

4. Simulate selective WhatsApp chat deletion

Create:

- chat.txt

Put 10 fake messages in it:

- 09:01 - Amit: Good morning
- 09:03 - Rahul: Are we meeting today?
- 09:05 - Amit: Yes, at 5 PM
- 09:10 - Rahul: deleted
- 09:15 - Neha: I will join too
- 09:20 - Amit: deleted
- 09:25 - Rahul: Okay
- 09:30 - Neha: See you there
- 09:35 - Amit: deleted
- 09:40 - Rahul: Bye

Now create:

- remove_deleted.py

Python script

```
input_file = "chat.txt"
```

```
output_file = "chat_cleaned.txt"
```

```
with open(input_file, "r", encoding="utf-8") as file:
```

```
    lines = file.readlines()
```

```
with open(output_file, "w", encoding="utf-8") as file:
```

```
    for line in lines:
```

```
        if "deleted" not in line.lower():
```

```
            file.write(line)
```

```
print("Lines containing 'deleted' have been removed.")
```

Run:

- python remove_deleted.py

The resulting chat_cleaned.txt will contain:

- 09:01 - Amit: Good morning
- 09:03 - Rahul: Are we meeting today?

- 09:05 - Amit: Yes, at 5 PM
- 09:15 - Neha: I will join too
- 09:25 - Rahul: Okay
- 09:30 - Neha: See you there
- 09:40 - Rahul: Bye

5. Detect possible steganography modification

Create:

- detect_stego.py

Use:

```
import os
from PIL import Image

def analyze_image(filename):
    size = os.path.getsize(filename)

    with Image.open(filename) as img:
        print(f"\nFile: {filename}")
        print(f"Format: {img.format}")
        print(f"Dimensions: {img.size}")
        print(f"File size: {size} bytes")
        print(f"Metadata fields: {len(img.getexif())}")

    return {
        "size": size,
        "format": img.format,
        "dimensions": img.size,
        "metadata": len(img.getexif())
    }
```

```
normal = analyze_image("normal.png")
stego = analyze_image("stego.png")

print("\n--- Comparison ---")

if normal["size"] != stego["size"]:
    print("File sizes are different.")
else:
    print("File sizes are identical.")

if normal["metadata"] != stego["metadata"]:
    print("Metadata differs.")
else:
    print("Metadata is the same.")

print("\nNote: These differences do NOT prove that steganography
was used.")
```

Install Pillow if necessary:

- pip install Pillow

Then put:

- normal.png
- stego.png
- detect_stego.py

in the same folder.

Run:

- python detect_stego.py